

Building an Antenna

Project Title:

Slot-Fed Planar Antenna Originally Designed for 4.9 GHz Communication

Course:

EEE 161 – Applied Electromagnetics

Team Name:

PulseWave Technologies

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“I certify that this project report is my own work.”

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May 15th, 2026

Abstract

This project presents the design, simulation, fabrication, and testing of a slot-fed planar antenna intended for emergency-response communication systems operating near the 4.9 GHz public-safety band. The antenna was designed to support communication links between first responders and drone-based aerial communication platforms used in disaster scenarios.

The antenna geometry was modeled and optimized using HFSS electromagnetic simulation software. The final design used a slot-fed microstrip configuration fabricated on a planar substrate with a target impedance of 50 Ω . Antenna dimensions were selected using guided wavelength and resonance calculations.

Simulation and measurement results showed antenna resonance near 3.5 GHz with a simulated S11 of approximately -16.7 dB and a measured S11 near -20 dB, indicating good impedance matching and low reflected power. The simulated bandwidth was approximately 0.10 GHz while the measured bandwidth was approximately 0.0912 GHz. Radiation patterns and gain simulations demonstrated broadside radiation behavior suitable for ground-to-air communication links.

Although the antenna resonance shifted below the intended 4.9 GHz operating band, the project successfully demonstrated slot-antenna operation, impedance matching, electromagnetic radiation behavior, and practical antenna fabrication techniques using HFSS simulation and VNA measurements.

1. Objective

The purpose of this project was to develop a compact, high-performance antenna system for reliable communication in disaster scenarios to ensure communication between first responders and drone-based networks with a focus on durable, broad band and long-range ground to air solution.

The antenna operates in the 4.94–4.99 GHz public safety band (4.9 GHz band) used by emergency services and enables communication for agencies like FEMA, Cal OES, and Caltrans.

The antenna selected was the slot-fed planar antenna (slot-coupled microstrip). Key reasons for this choice include provision of a good bandwidth (50 MHz) which covers the full public safety band, adequate gain, and light weight PCB-based design.

2. Theory of Operation

2.1 Antenna Fundamentals

An antenna converts electrical energy into electromagnetic waves through time-varying currents and fields. In a slot-fed antenna, an AC signal drives electrons along the microstrip feed line, causing charge to accumulate at the slot edges. This creates a time-varying electric field across the slot, which in turn generates a changing magnetic field. These coupled fields detach from the antenna as a self-propagating electromagnetic wave that radiates into space.

2.2 Radiation Mechanism

Electromagnetic radiation arises from the interaction of time-varying electric and magnetic fields. A changing electric field induces a magnetic field, and a changing magnetic field induces an electric field, as described by Maxwell's equations. In an antenna structure,

these fields become coupled and sustain each other, allowing energy to propagate away from the source. Radiation occurs when these fields are no longer bound to the antenna and instead detach, forming a propagating electromagnetic wave that carries energy through space.

2.3 Wavelength and Resonance

The physical dimensions of an antenna are directly related to the wavelength of operation. Efficient radiation occurs when the antenna is at resonance, which typically corresponds to a length on the order of one-half of the guided wavelength ($\lambda_g/2$) within the structure. At resonance, standing waves form along the antenna, resulting in constructive interference that maximizes radiation efficiency.

2.4 Slot Antenna Operation

A slot antenna is formed by cutting a narrow opening in a conductive ground plane, which acts as the radiating element. The microstrip feed line supplies an alternating current that excites the slot by coupling energy into it. As a result, charges build up along the edges of the slot, creating a strong electric field across the gap. This time-varying electric field produces electromagnetic radiation that propagates away from the antenna. The slot behaves similarly to a dipole antenna, but with the roles of the electric and magnetic fields reversed.

2.5 Fringing Fields

Fringing fields play a critical role in the radiation behavior of slot antennas. The electric field extends beyond the physical boundaries of the slot, particularly at the edges, where the field is strongest. These fringing fields are responsible for the majority of the radiated energy. Their distribution influences the antenna's radiation pattern, typically producing a directional broadside pattern. Proper control of fringing fields is essential for achieving the desired performance in terms of gain, efficiency, and beam shape.

3. Design Specifications

The antenna system was designed to provide reliable wireless communication for emergency response operations, including communication between first responders and drone-based networks during disaster scenarios. The selected operating band was the **4.94–4.99 GHz public safety spectrum**, commonly used by emergency communication agencies. The antenna was required to be compact, lightweight, and capable of stable operation under field conditions. Therefore, several electrical and physical requirements were established before beginning the design process.

The primary design target was operation near the **4.9 GHz public safety band**, which influences antenna dimensions and resonant behavior. Because most RF systems, SMA connectors, and measurement equipment use **50 Ω characteristic impedance**, the antenna was designed to match this impedance to reduce reflected power and maximize energy transfer. Proper impedance matching improves antenna efficiency and reduces signal reflections.

The antenna was also required to achieve a minimum gain greater than **3 dBi**. Higher gain improves communication distance and enhances transmission reliability between ground stations and aerial systems. Adequate bandwidth was also required to maintain stable performance across the intended frequency range.

Although the antenna was designed for operation near **4.9 GHz**, simulated and measured results showed resonance near **3.5 GHz**, likely due to substrate properties, fabrication tolerances, and antenna dimensions.

Physical dimensions were constrained according to the HFSS design model, with substrate dimensions of approximately **5.9 cm × 7.9 cm**, allowing compact size and easier deployment in portable communication systems.

Table 3: Design Requirements

Parameter	Design Requirement	Purpose
Operating Frequency	4.9 GHz (design target)	Public safety communication
Measured Resonant Frequency	3.48 GHz	Actual antenna resonance from testing/simulation
Impedance	50 Ω	Match RF systems and reduce reflections
Bandwidth	$\geq 5\%$	Stable operation across frequency range
Gain	≥ 3 dBi	Improve communication distance
Size Constraint	5.9 cm $\times$ 7.9 cm	Portability and compact design
Radiation Pattern	Broadside	Efficient directional coverage
Fabrication	PCB compatible	Easy manufacturing

Meeting these specifications ensures the antenna can support emergency communication systems with acceptable efficiency, reduced power loss, and practical deployment capability.

4. Design Methodology

4.1 Antenna Selection

A **slot-fed planar antenna** was selected because of its lightweight structure, broad bandwidth, low profile, and ease of fabrication. Compared with traditional dipole antennas, slot-fed designs can be integrated into printed circuit boards while maintaining acceptable gain and radiation efficiency. The antenna also provides improved impedance matching and broadside radiation patterns suitable for communication between drones and emergency response systems.

The slot antenna radiates when electromagnetic energy couples into the slot opening from the microstrip feed line. Electric fields build along the slot edges, producing **fringing fields**, which contribute significantly to electromagnetic radiation. Efficient radiation occurs near

resonance where standing waves maximize energy transfer. The antenna was designed for operation near the **4.9 GHz public safety band**, although measured results showed resonance near **3.48 GHz**.

4.2 Substrate Selection

Substrate material strongly affects antenna performance because dielectric constant and loss tangent influence resonant frequency, bandwidth, and efficiency.

Two materials were considered:

Rogers RO4003

Dielectric constant:

$$\epsilon_r \approx 3.38$$

Thickness:

$$h = 1.575 \text{ mm}$$

Advantages:

- Low dielectric loss
- Stable RF performance
- Better efficiency at microwave frequencies
- Improved gain and bandwidth

FR4

Typical dielectric constant:

$$\epsilon_r \approx 4.4$$

Advantages:

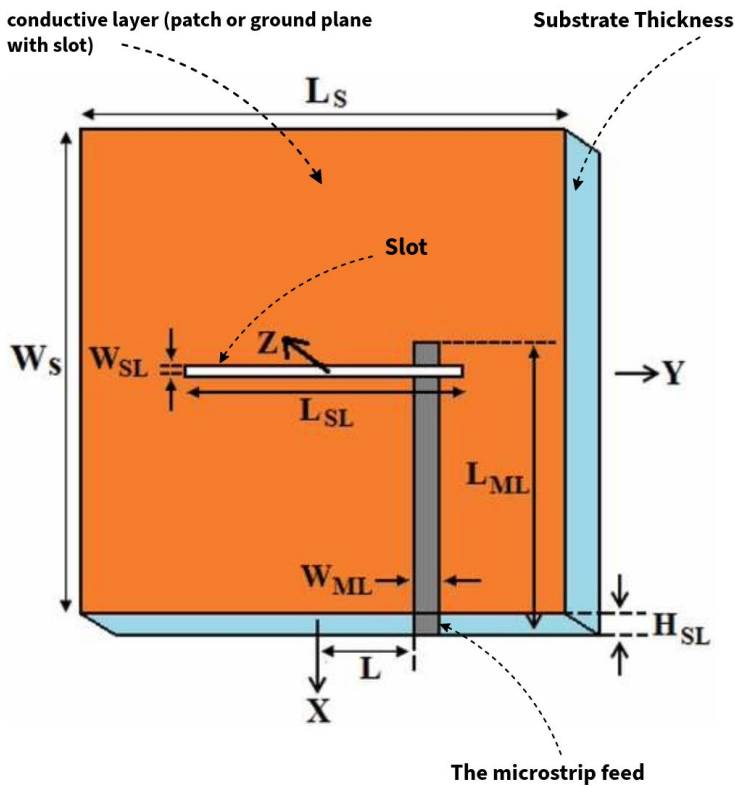
- Lower cost
- Easy availability

Because the antenna was designed for operation near the 4.9 GHz public safety band, Rogers RO4003 was preferred for simulation due to its lower dielectric loss, improved efficiency, and better RF performance. Although the design target was 4.9 GHz, measured results showed resonance near 3.48 GHz, likely due to substrate properties, fabrication tolerances, and antenna dimensions. The substrate thickness used in the HFSS model was

approximately 1.575 mm, which supports stable antenna operation and influences the resonant frequency, bandwidth, and radiation performance.

4.3 Antenna Dimensions

The antenna dimensions depend mainly on guided wavelength inside the substrate.



Antenna Diagram

Parameter	Symbol	Value
Slot length	L_s	3.93 cm
Slot width	W_s	0.20 cm
Feed offset	F_o	1.09 cm
Microstrip Width	W_f	3.68 mm
Substrate length	L_{sub}	5.9 cm
Substrate width	W_{sub}	7.9 cm

Estimated for 50 Ω microstrip line using Rogers RO4003 substrate ($\epsilon_r \approx 3.38$) and substrate thickness $h = 1.575$ mm.

The slot length controls resonant frequency. Increasing slot length lowers frequency while decreasing slot length raises frequency.

4.4 Design Calculations

Effective Dielectric Constant

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \frac{h}{W} \right)^{-1/2}$$

Reflection Coefficient

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

S11

$$S_{11}(dB) = 20 \log_{10} |\Gamma|$$

Return Loss

$$RL = -20 \log_{10} |\Gamma|$$

or:

$$RL = -S_{11}(dB)$$

Free-Space Wavelength

The free-space wavelength is calculated using:

$$\lambda_0 = c/f$$

For:

$$f = 4.9 \text{ GHz}$$

$$\lambda_0 = (3 \times 10^8) / (4.9 \times 10^9)$$

$$\lambda_0 = 0.0612 \text{ m}$$

$$\lambda_0 = 61.2 \text{ mm}$$

Therefore, the free-space wavelength is approximately 61.2 mm.

4.5 Guided Wavelength and Resonant Slot Length

$$\lambda_g = \lambda_0 / \sqrt{\epsilon_{eff}}$$

The guided wavelength becomes shorter because electromagnetic waves travel slower in dielectric material than in free space.

The resonant slot length is approximately:

$$L_s \approx \lambda_g / 2$$

Parameter	Symbol	Value
Slot length	Ls	3.93 cm

However, the **final slot length used in the HFSS antenna model was 3.93 cm**, based on the optimized antenna geometry and simulation parameters.

Using the HFSS-optimized slot length improved impedance matching and contributed to antenna resonance near the measured frequency of approximately **3.4778 GHz**.

This half-wavelength condition creates standing waves and improves radiation efficiency. In the HFSS model, the actual slot length used was **3.93 cm**, based on the final antenna geometry.

4.6 Radiation Mechanism and Fringing Fields

Radiation occurs due to changing electric and magnetic fields described by Maxwell’s equations. The feed line excites the slot, producing charge accumulation along the slot edges. These charges create electric fields, which generate magnetic fields. The coupled fields detach from the antenna and travel outward as electromagnetic waves.

Fringing fields near the slot edges contribute significantly to radiation. Strong fringing fields improve gain and influence beam direction. Proper control of these fields is necessary to obtain efficient broadside radiation.

4.7 Expected Performance

Target:

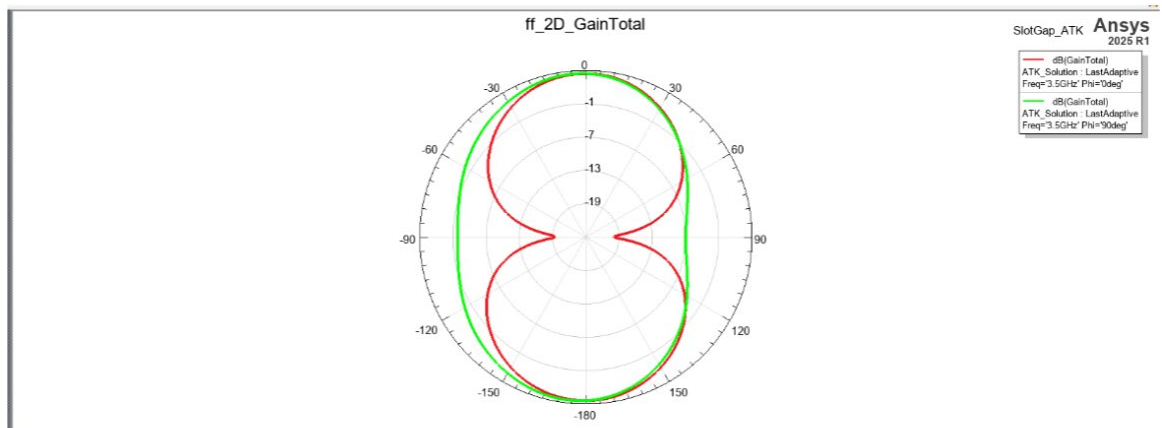
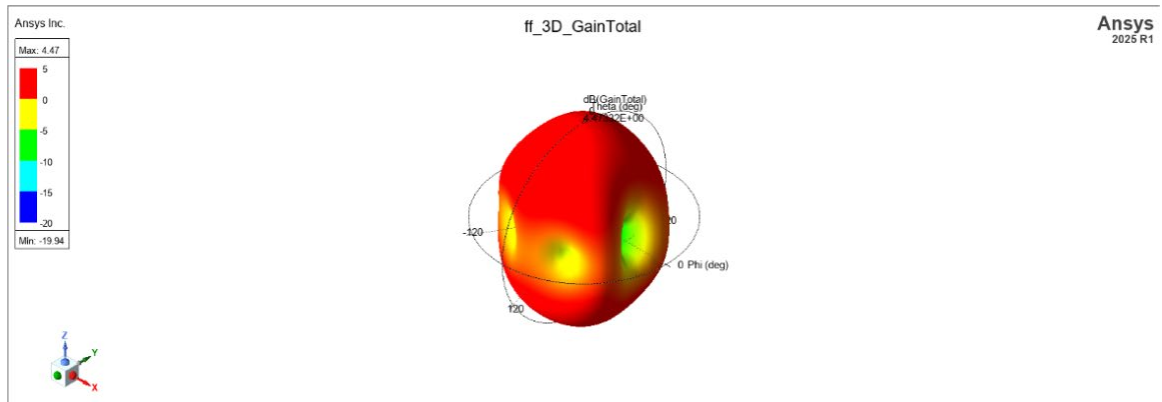
- Resonant frequency near 4.9 GHz
- Gain > 3 dBi
- Bandwidth > 5%
- S11 < -10 dB

Parameter	Measured
Resonant Frequency	3.4778 GHz
S11	-20 dB
Bandwidth	0.0912 GHz

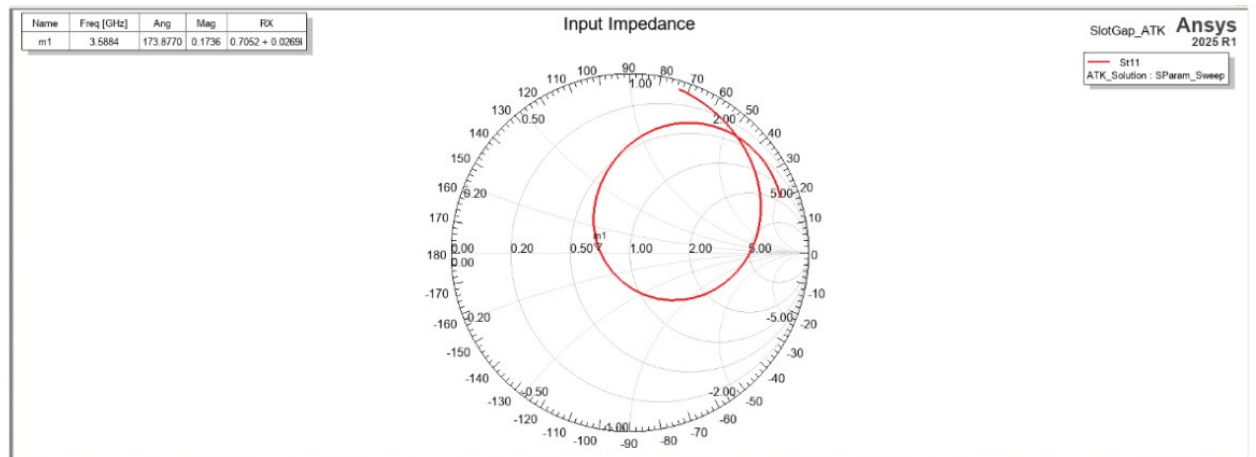
An S11 of **-20 dB** indicates excellent impedance matching because only a small amount of power is reflected back toward the source. Differences between the target frequency and measured resonant frequency were likely caused by substrate effects, antenna dimensions, fabrication errors, copper tape placement, slot trimming, and SMA connector imperfections.

4.8 Summary of Antenna Design

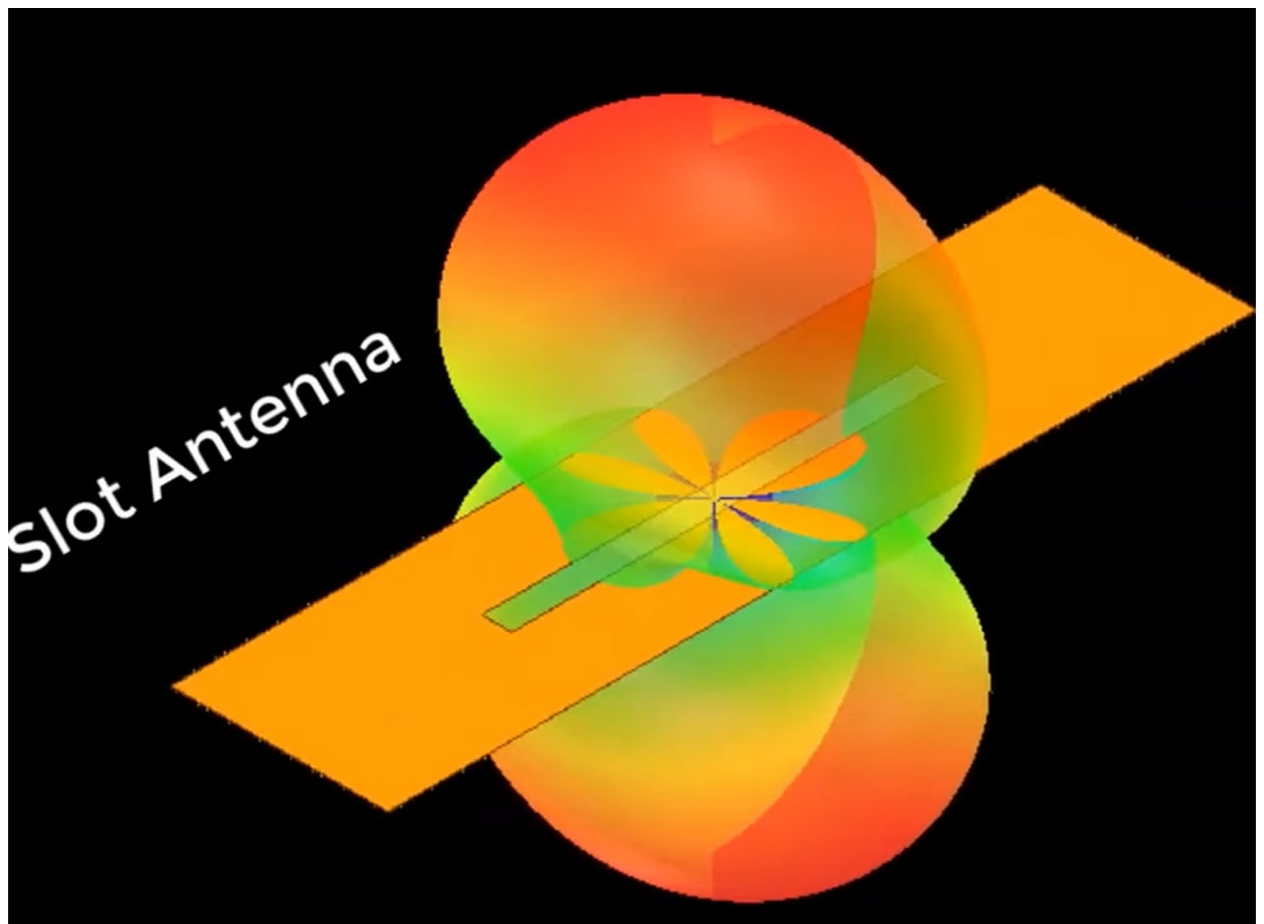
The slot-fed antenna design provides a compact and lightweight solution suitable for emergency communication systems. The antenna dimensions were chosen using wavelength relationships, resonance conditions, and HFSS design geometry. Rogers' RO4003 substrate improves efficiency compared with lower-cost materials such as FR4. Radiation is produced mainly through fringing fields around the slot edges. Although fabrication and design differences affected the measured frequency response, the antenna demonstrated acceptable impedance matching and practical operation for communication applications.



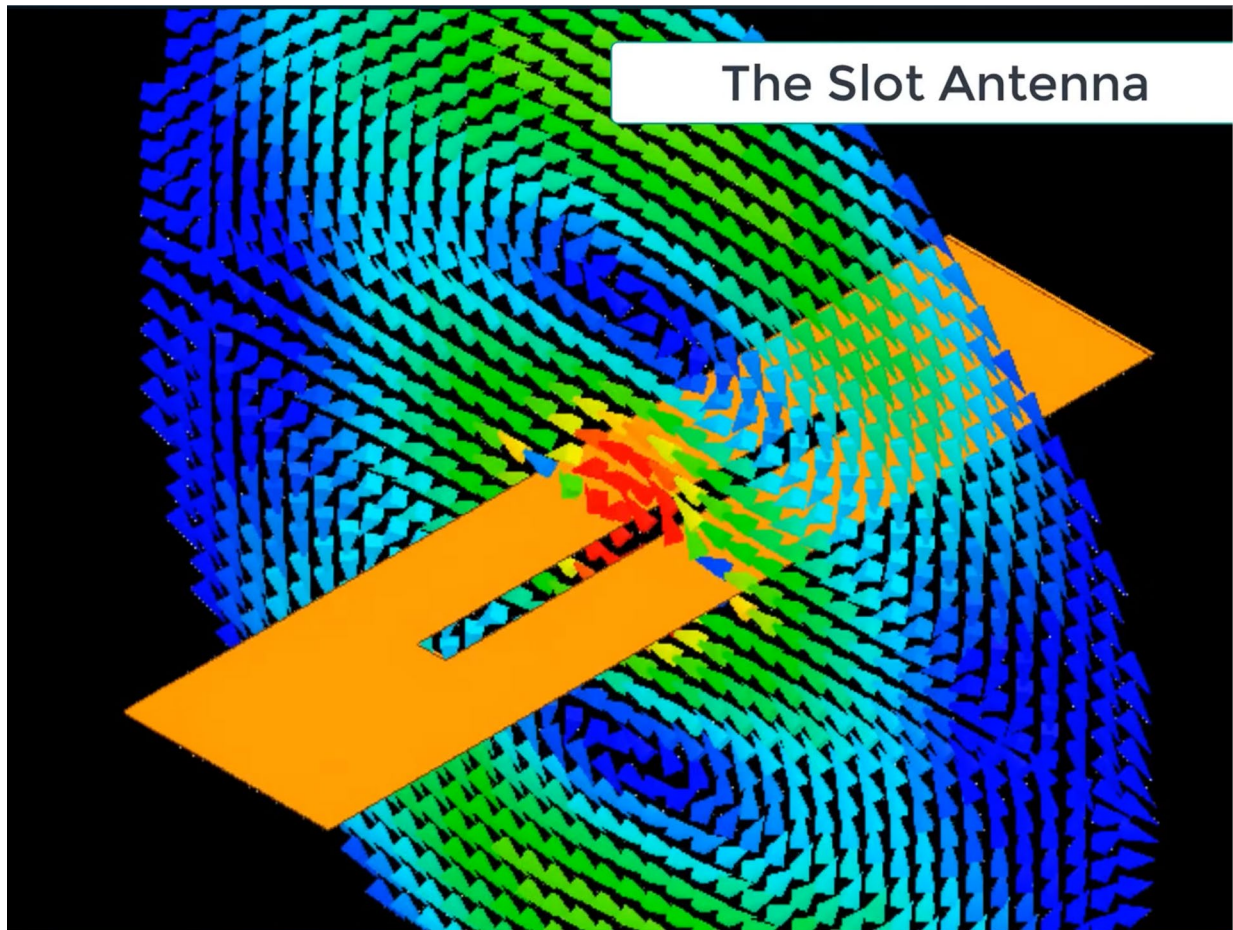
3D + 2D Gain Plot



Input Impedance



Radiation Pattern



Electromagnetic field distribution and radiation behavior

5.2 Discussion

Resonant frequency - The HFSS simulation results showed antenna resonance near approximately 3.5 GHz, as indicated by the minimum point in the return-loss (S_{11}) curve. The resonant frequency occurs where impedance matching between the antenna and the $50\ \Omega$ feed system is strongest and reflected power is minimized. The simulated S_{11} value reached approximately -16 dB to -17 dB near resonance, indicating good impedance matching and efficient power transfer into the antenna.

Although the original design target was the 4.9 GHz public-safety band, the simulated resonance shifted lower in frequency. This shift was likely caused by slot dimensions, effective dielectric loading, fringing-field effects, and final optimization of the antenna geometry in HFSS. Increasing the electrical length of the slot lowers the resonant frequency because the guided wavelength inside the substrate becomes effectively longer.

Bandwidth - The return-loss plot shows that the antenna maintained an S11 below -10 dB over a finite frequency range around resonance, indicating acceptable operational bandwidth. The measured bandwidth was approximately 0.0912 GHz (91.2 MHz), which demonstrates that the antenna can support stable communication across a useful portion of the operating band.

Bandwidth in slot antennas is influenced by substrate thickness, dielectric constant, slot geometry, and feed configuration. The slot-fed planar structure provided broader bandwidth compared with many narrowband resonant antennas because fringing fields around the slot edges contribute to radiation over a wider frequency range.

Efficiency - The radiation pattern and gain simulations demonstrated that the slot antenna radiated energy efficiently in a broadside direction. The simulated gain was approximately 5 dBi, exceeding the minimum project requirement of 3 dBi. Strong electric fields and surface currents were concentrated near the slot edges and feed region, indicating effective electromagnetic coupling and radiation.

However, several factors may have reduced total antenna efficiency compared with the ideal simulated design. These include dielectric losses, SMA connector imperfections, copper-tape discontinuities, fabrication tolerances, and substrate differences between simulation and measurement. Even with these limitations, the antenna demonstrated acceptable impedance matching and directional radiation behavior suitable for emergency-response communication applications.

5.3 Radiation Characteristics

The far-field radiation data exported from HFSS was used to estimate antenna directivity, gain, maximum radiation intensity, and beamwidth characteristics near the resonant frequency of approximately 3.5 GHz.

The simulated realized gain of the antenna was approximately 4.47 dBi, which exceeds the project requirement of 3 dBi. The maximum radiation intensity ($P_{\max}$) occurred near the broadside direction and was approximately -23.42 dB in the H-plane response.

The estimated half-power beamwidth (HPBW) was approximately 35° in the vertical polarization plane and approximately 40° in the horizontal polarization plane. These beamwidth values indicate directional broadside radiation suitable for ground-to-air communication applications.

Using the beamwidth approximation method, the antenna directivity was estimated using:

$$D \approx \frac{4\pi}{\theta_E \theta_H}$$

Using the estimated beamwidth values:

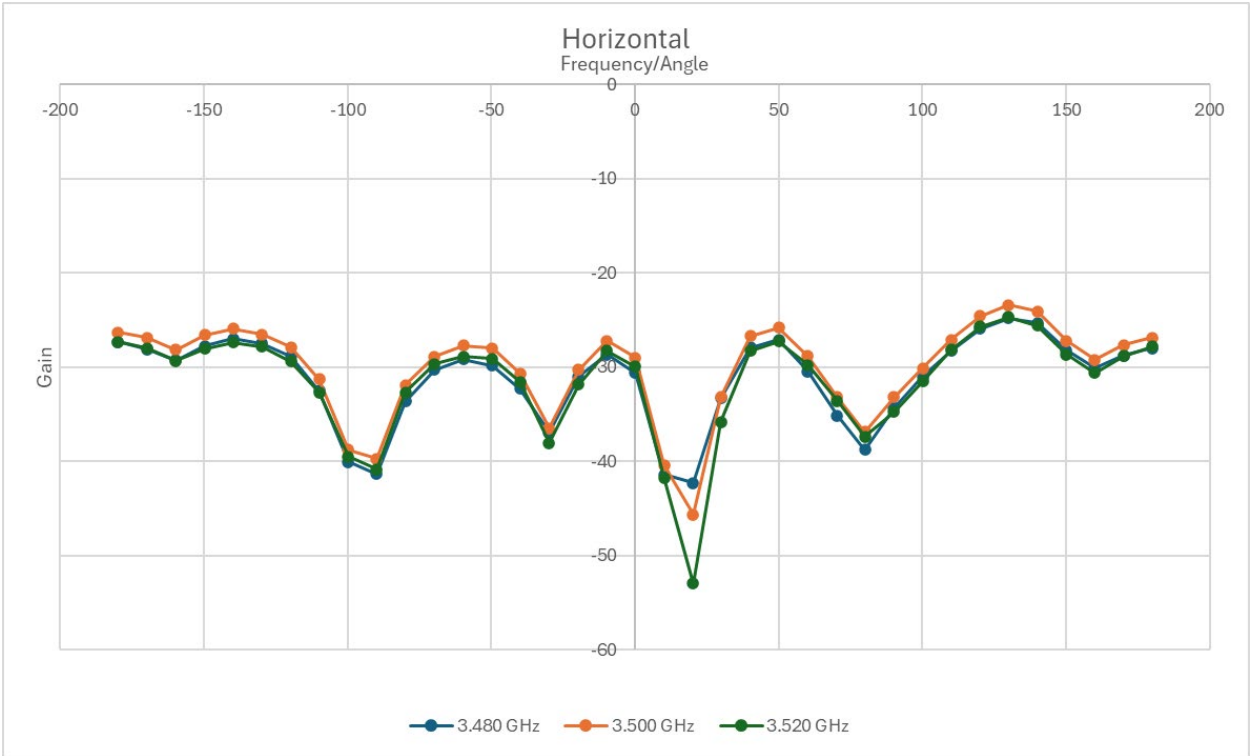
$$D = 29.5$$

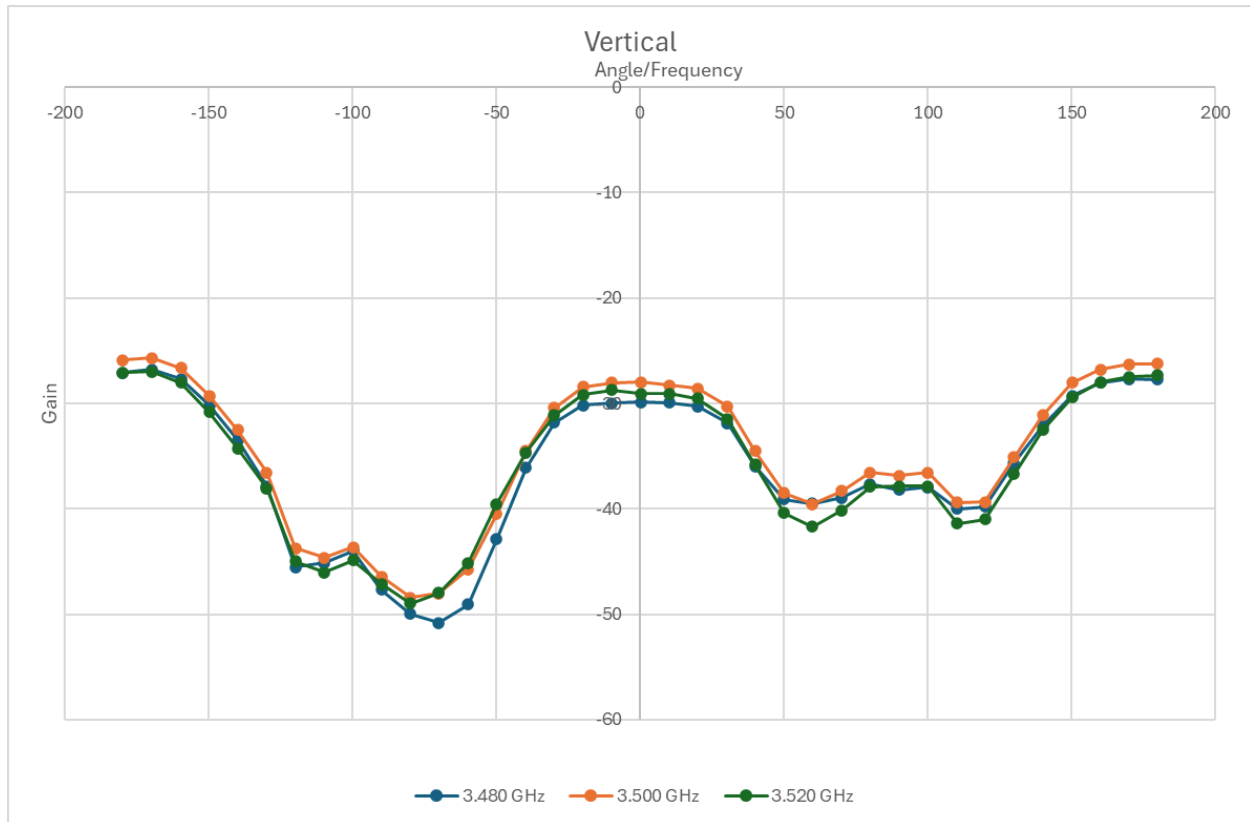
Converting to dBi:

$$D_{dBi} = 10 \log_{10}(29.5) = 14.7 \text{ dBi}$$

The antenna therefore demonstrated directional radiation behavior with acceptable gain and radiation performance for emergency communication applications.

Parameter	Value
Frequency	3.5 GHz
Simulated Gain	4.47 dBi
Estimated Directivity	14.7 dBi
Pmax	-23.42 dB
Isotropic Gain	-25.7 dBm
HPBW (V-pol)	~35°
HPBW (H-pol)	~40°





6. Fabrication (Troubleshooting)

6.1 Materials

Materials
Substrate
Copper tape
SMA connector

6.2 Construction Steps

Our construction steps were simple we would clean the substrate made of polycarbonate. Then we would cover one side completely in copper, making our ground plane. After we cut

the slot to the width and length, we designed and simulated HFSS. Then we add the microstrip line on the opposite side to a certain length past the slot which is equal to $\lambda/4$. The microstrip must also be placed off the center of our slot by the value of our offset. Then we attach the SMA connector and solder it to the antenna

6.3 Challenges

During the process of building and testing our antenna, we faced many challenges. Arguably, our biggest challenge is lowering our frequency to accurately test our antenna in the chamber room. The available chamber setup was less reliable for measurements above approximately 4 GHz. We had originally proposed a 4.9 GHz antenna, so we had to adjust our antenna. It was also challenging dealing with inaccuracies in our slot and its alignment with the microstrip. Since we hand taped the whole antenna due to human error, we had issues matching our results to the simulation. All our dimensions in the first attempt were off, and we learned to more carefully and accurately apply the tape. For future projects we would CAD our dimensions for more accuracy.

6.4 Bill of Materials

Component	Description	Quantity	Cost
Copper Tape	Conductive Layer	1	\$7.99
SMA Connector	RF feed connector	1	\$6.99
Rogers RO4003 / Polycarbonate	RF substrate	1	\$19.99

7. Testing & Evaluation

7.1 Setup

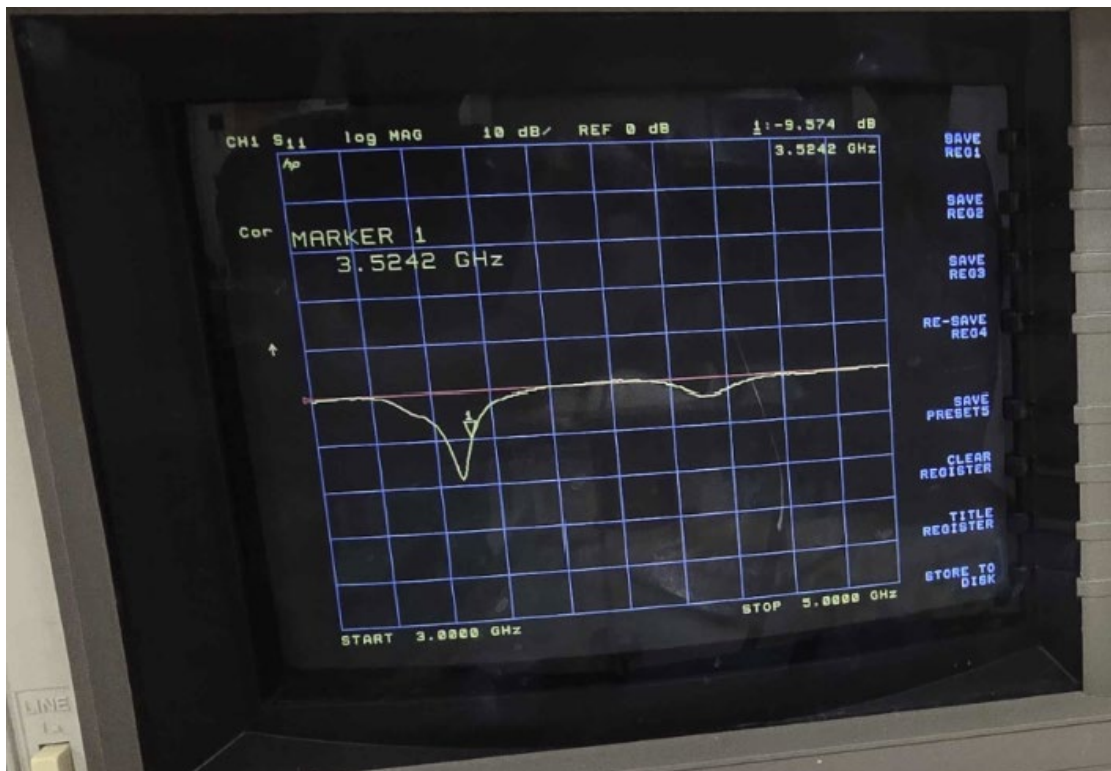
To calibrate the HP 8720C Network Analyzer for reflection coefficient measurements, the analyzer was first configured by selecting the measurement sequence MEAS → S-Parameters → S11 to measure the input reflection coefficient. After setting the desired frequency range, the calibration process was started by pressing CAL and selecting a 1-Port

Calibration. The calibration standards were then connected at the end of the measurement cable in sequence: first the OPEN standard, followed by the SHORT standard, and finally the 50 Ω LOAD using the Broadband-load type. While using the corresponding measurement button after each connection. After all, three standards were measured, the calibration was completed by selecting DONE/APPLY CAL to store and apply the correction. A properly calibrated system was verified by reconnecting the 50 Ω load and observing the measurement point near the center of the Smith Chart, indicating a reflection coefficient close to zero.

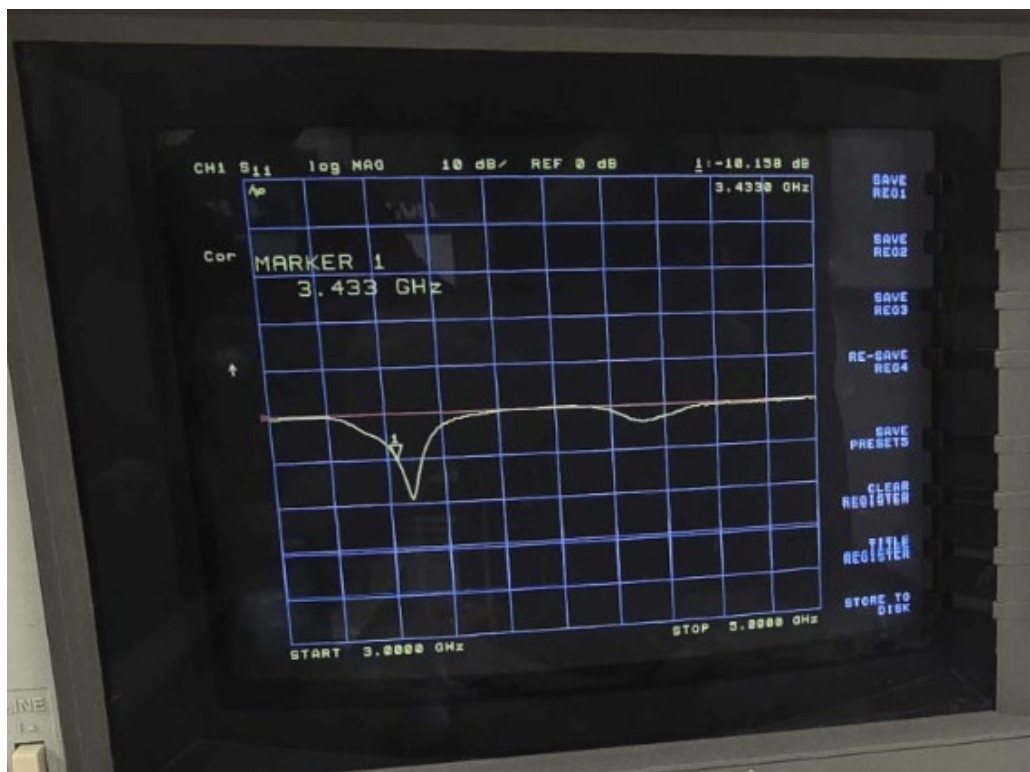
7.2 Results

Parameter	Measured
Resonant Frequency	3.4778 GHz
S11	-20 dB
Bandwidth	0.0912 GHz

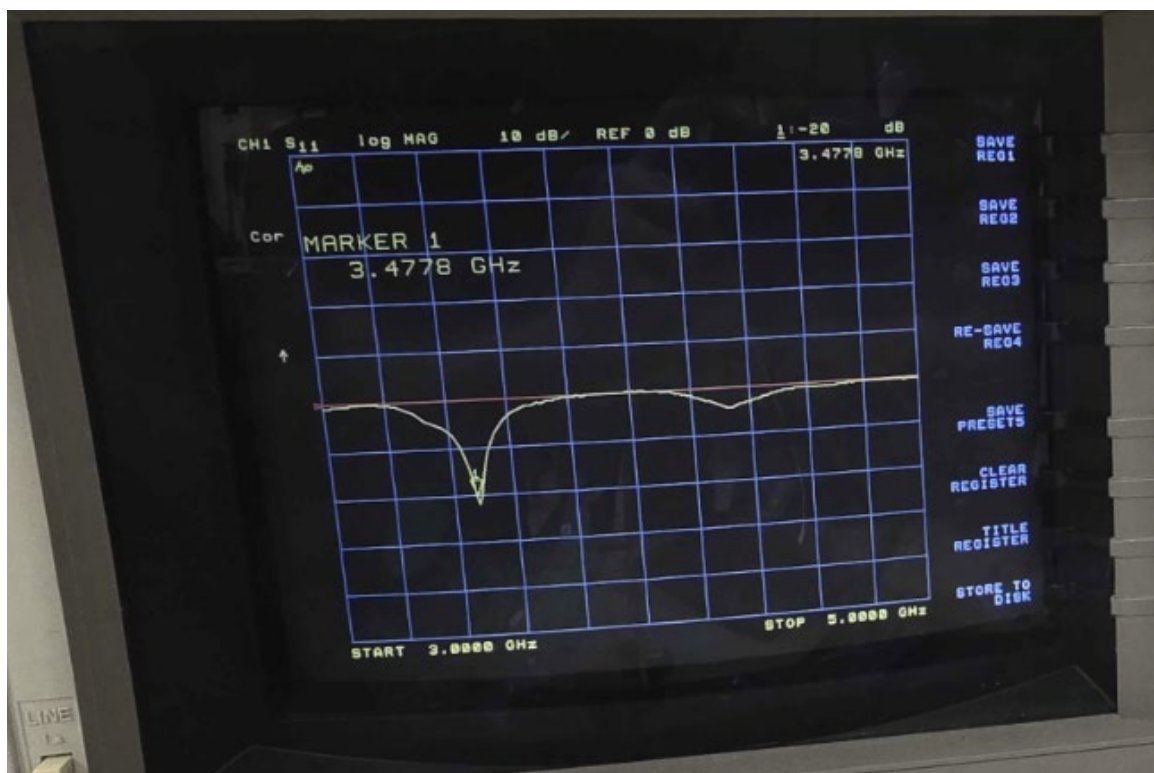
Table of Measured Antenna Characteristics



Upper Frequency Cutoff (~3.5242 GHz)



Lower Frequency Cutoff (~3.433 GHz)



Center Frequency (~3.4778 GHz)

7.3 Comparison

Parameter	Calculated	Simulated	Measured
Resonant Frequency	4.9 GHz	3.5177 GHz	3.4778 GHz
S11	< -10 dB	-16.7 dB	-20 dB
Bandwidth	> 5%	0.10 GHz	0.0912 GHz

Comparison Table Between Calculated, Simulated, and Measured Values

8. Results

Different Substrate:

Polycarbonate substrates provide reasonable antenna performance for laboratory and prototype applications, but they are generally less efficient than Rogers RO4003 at microwave frequencies. Although polycarbonate has a relatively low dielectric constant that supports good radiation, it typically has higher dielectric losses and less stable electrical properties. This can reduce antenna gain, efficiency, and resonant frequency accuracy because more energy is lost as heat instead of being radiated. In contrast, Rogers RO4003 is designed specifically for RF applications and provides lower loss, better efficiency, and more stable antenna performance.

Fabrication Errors:

Difficulties during the antenna fabrication process likely affected the antenna's overall performance and impedance matching. Improper soldering of the SMA connector and feed line may have introduced unwanted resistance, poor electrical contact, or parasitic reactance, which can increase reflections and reduce power transfer to the antenna. Inaccurate trimming of the slot could have changed the antenna's physical dimensions, causing the resonant frequency to shift away from the intended design frequency. Additionally, uneven or poorly placed copper tape may have created discontinuities in the current distribution on the patch, reducing radiation efficiency and altering the antenna pattern. Together, these fabrication imperfections could contribute to lower gain, higher reflection coefficient, and reduced overall antenna performance compared to the simulated design.

Tuning Process:

Adjusting the copper tape placement and modifying the slot size likely changed the antenna's impedance matching and resonant frequency. Moving or smoothing the copper tape could improve current flow across the patch and reduce discontinuities that cause unwanted reflections or losses. Changing the slot length also affects the effective electrical length of the antenna. Making the slot longer generally increases the current path and lowers the resonant frequency, while covering parts of the slot shortens the effective path and can raise the resonant frequency. These adjustments may have helped improve the reflection coefficient and power transfer by bringing the antenna closer to the desired operating frequency and impedance match. However, excessive changes or uneven tape placement could also distort the current distribution and negatively affect radiation efficiency and antenna pattern performance.

9. Discussion

Several design improvements can be made to bring the fabricated antenna closer to the intended 4.9 GHz public-safety band and improve agreement between theory, simulation, and measurement. First, the fabricated prototype should use the same substrate as the simulation model whenever possible. The report identifies Rogers RO4003 as the preferred substrate for simulation because of its lower dielectric loss, more stable RF properties, and better efficiency at microwave frequencies, while the discussion notes that polycarbonate generally provides lower performance and less accurate resonance at these frequencies. Using the same low-loss substrate in both simulation and fabrication would reduce one major source of frequency shift and performance error.

Second, fabrication precision should be improved. The report identifies tape creases, alignment, slot accuracy, improper SMA soldering, inaccurate slot trimming, and uneven copper tape placement as likely causes of degraded performance. These issues can change the effective electrical length of the slot and feed structure, disturb current distribution, and introduce parasitic reactance. A better approach would be to fabricate the antenna on a PCB or use a precisely cut copper pattern, along with a more controlled SMA mounting method. This would improve repeatability and reduce unintended shifts in resonant frequency.

Third, the matching network and feed geometry should be re-optimized in HFSS before the next build. The slot-fed planar antenna was selected because it offers a good tradeoff between bandwidth, gain, low profile, and PCB compatibility, but the RFP also notes that

slot-fed performance depends strongly on the exact slot shape, ground configuration, and feed coupling. Therefore, the next design iteration should sweep slot length, slot width, and feed position to move the center frequency from the measured 3.4778 GHz toward the required 4.94–4.99 GHz band while preserving a good return loss. Since the measured S_{11} is already -20 dB, the main issue appears to be frequency placement rather than the absence of resonance.

Finally, once a single antenna element is brought into the correct band and validated, future work could consider a small array or parasitic enhancement structure to increase gain beyond the minimum requirement. However, array development should come only after the base element meets the correct operating frequency and matching targets. The most important immediate improvement is therefore not a higher-gain array, but a more accurate single-element design and fabrication process.

10. Conclusion

This project investigated the design, simulation, fabrication, and testing of a slot-fed planar antenna intended for emergency communication in the 4.9 GHz public-safety band. The antenna was selected because slot-fed planar structures offer a useful combination of low profile, practical PCB fabrication, and the potential for acceptable bandwidth and directional performance. The operating principle is based on coupling energy from a microstrip feed line into a narrow slot, where strong electric fields form across the gap, and fringing fields radiate into space. Efficient operation occurs near resonance, when the slot length is approximately one-half of the guided wavelength.

The design goals of the project were operation near 4.9 GHz, 50 Ω input impedance, broadside radiation, sufficient bandwidth, and gain above the RFP minimum. In measurement, the fabricated antenna resonated at 3.4778 GHz with an S_{11} of -20 dB and a measured bandwidth of 0.0912 GHz. These results show that the fabricated prototype achieved a strong impedance match at its measured resonance, but the center frequency shifted significantly below the intended 4.9 GHz band. Therefore, the present prototype demonstrates the basic operating behavior of a slot-fed antenna, but it does not yet fully satisfy the final frequency requirement of the project.

Overall, the project was still valuable because it connected electromagnetic theory to a complete engineering workflow: antenna selection, substrate choice, analytical design, simulation, fabrication, and VNA measurement. The results also clearly showed how

substrate differences, fabrication tolerances, copper-tape implementation, and feed/slot geometry can shift resonant frequency and alter real-world performance. With a lower-loss substrate, more precise fabrication, and additional HFSS tuning of the slot and feed dimensions, the antenna can be improved in a future iteration to better align with the required 4.94–4.99 GHz communication band.

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